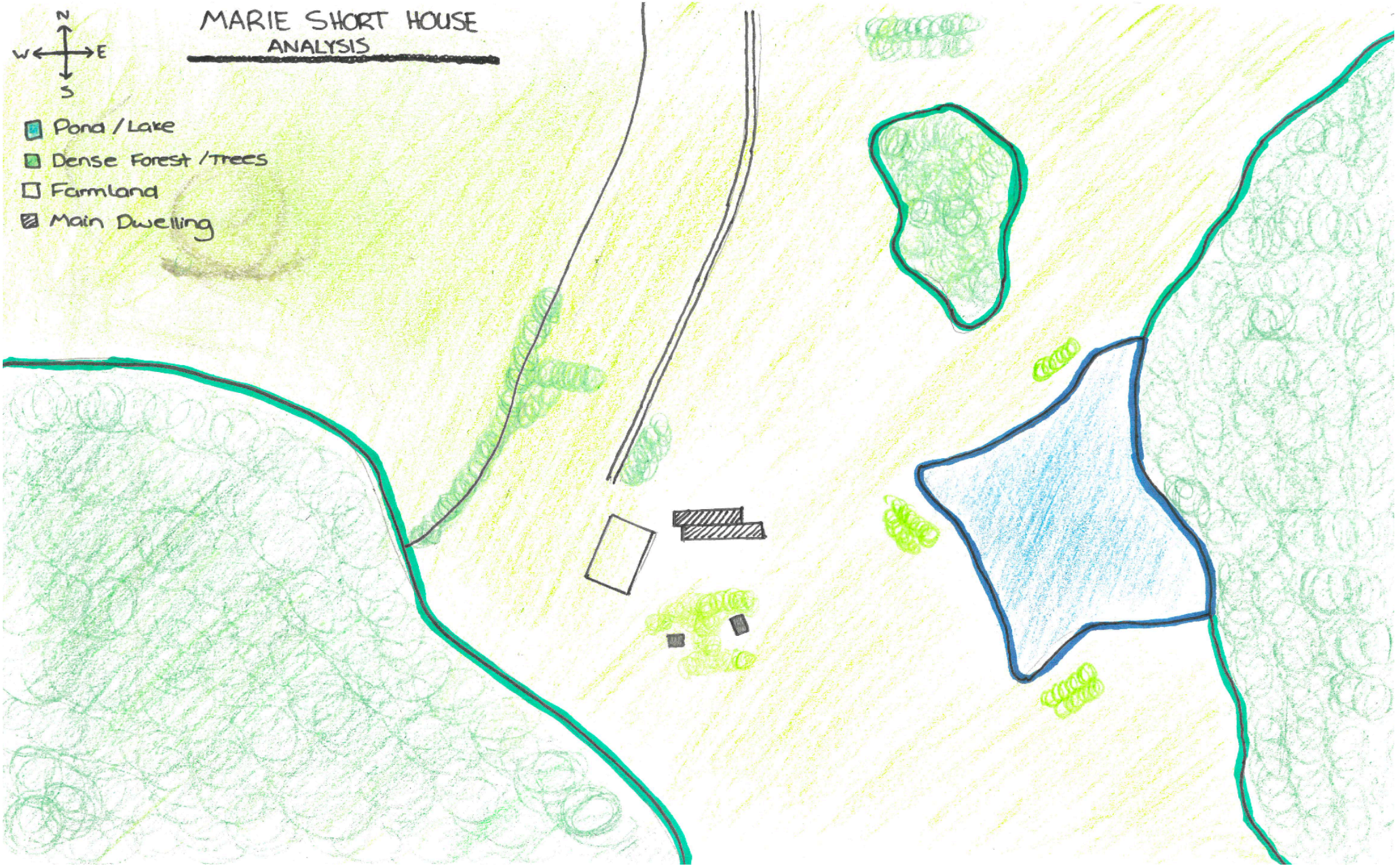
Portfolio A and B – Marie Short
House designed by 'Glenn Murcutt'

Presented by Fergus Shaw

Analysis Diagrams

- A draft analysis drawing I completed prior to beginning on photoshop.



MARIE SHORT HOUSE DESIGNED BY GLENN MURCUTT

The location of the Marie Short house is situated on a rural farm land. The pond located on the eastern side of the property allows for cooler winds to be blown in, therefore during summer months cools the house. The large foresty trees to the southern side reduce the amounts of wind blown in during winter months, allowing the house to stay warmer.

The louvres seen in the photograph above allow for cross venitlation from winds coming north - south.

Property and planning overlays:
- RU1 (Primary Production)
- RU2 (Rural Landscape)
- BMO (Bushfire Prone Land)
- HO (Heritage Overlay)
- VPO (Vegetation Prone Overlay)

Site Details:
- Latitude: 31°04S,
- Longitude: 152°50E
- Climate Zone: Humid subtropical climate
- Total Area: 3,308km²
- Elevation: approx 10m above sea level

Vegetation:
The foresty vegetation surrounding the property allow for natural habitats for local wild life.

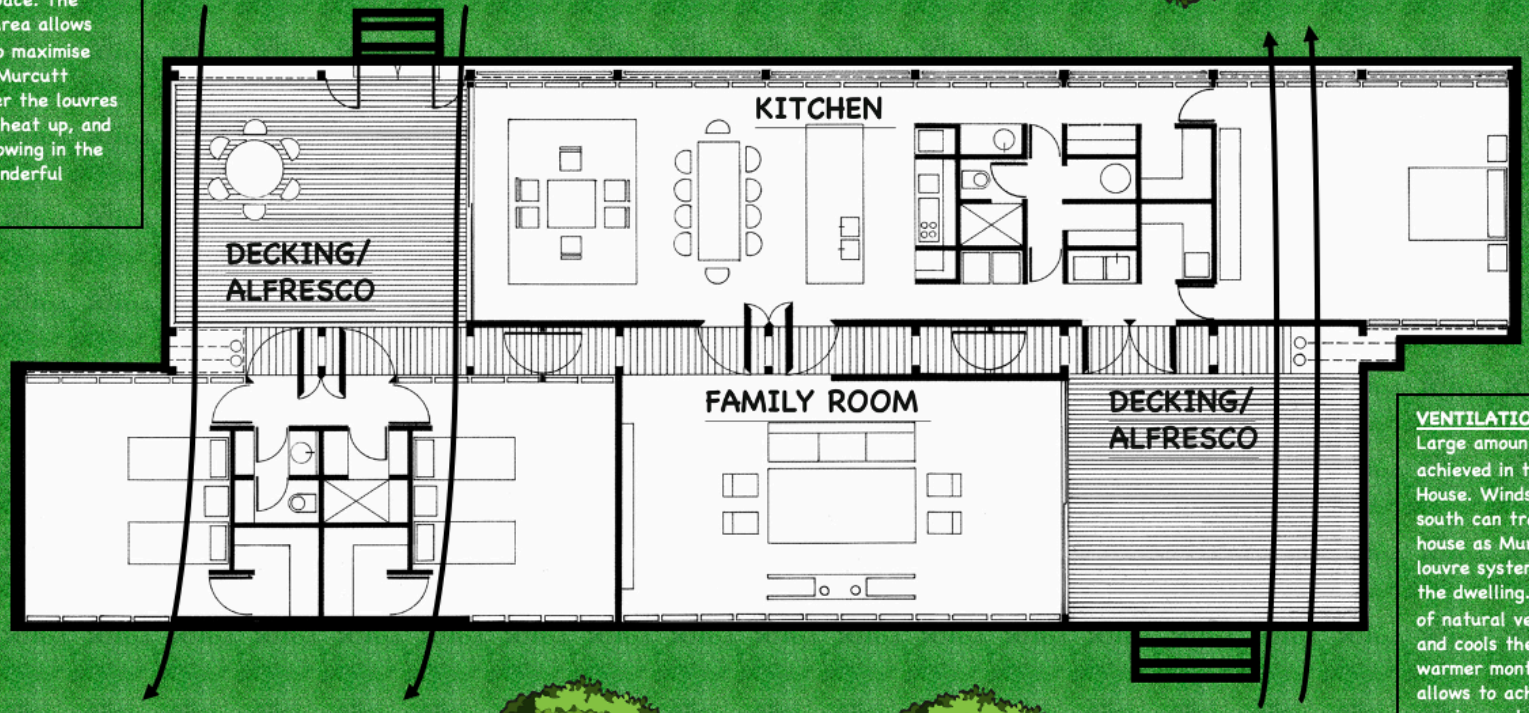
Also, the bushfire overlay indicated above, suggests due to the large amounts of trees that it may be a BAL 19 to BAL 40. Therefore, the Murcutt would have needed to design the house to these measures.

SUN PATH DETAILS:
Murcutt designed the house to a specific orientation. The living pavillion faces north to recieve sunlight for majority of the day. North facing windows haver been specifically placed to attract the warm sunlight. Furthermore, has created open farm land to the north to ensure no sunlight is blocked from large trees.

PORTFOLIO PART B – FLOOR PLAN ANALYSIS

MARIE SHORT HOUSE – FLOOR PLAN

Murcutts Design: Glenn Murcutt designed the house around the concept of maximising the sunlight from the north. This has been achieved by large windows and a louvre system to the north kitchen and living space. The decking / alfresco area allows for a great space to maximise sunlight and relax. Murcutt quotes "in the winter the louvres have a tendency to heat up, and when the wind is blowing in the summer, it has a wonderful cooling effect".



VENTILATION (black arrows): Large amounts of ventilation is achieved in the Marie Short House. Winds from the north and south can travel through the house as Murcutt has installed a louvre system on both sides of the dwelling. This allows plenty of natural ventilation to occur and cools the house down during warmer months. Furthermore, allows to achieve the NCC requirement of a minimum of 5% ventilation.

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PHOTOSHOP - REGIONAL ANALYSIS

MARIE SHORT HOUSE - REGIONAL ANALYSIS

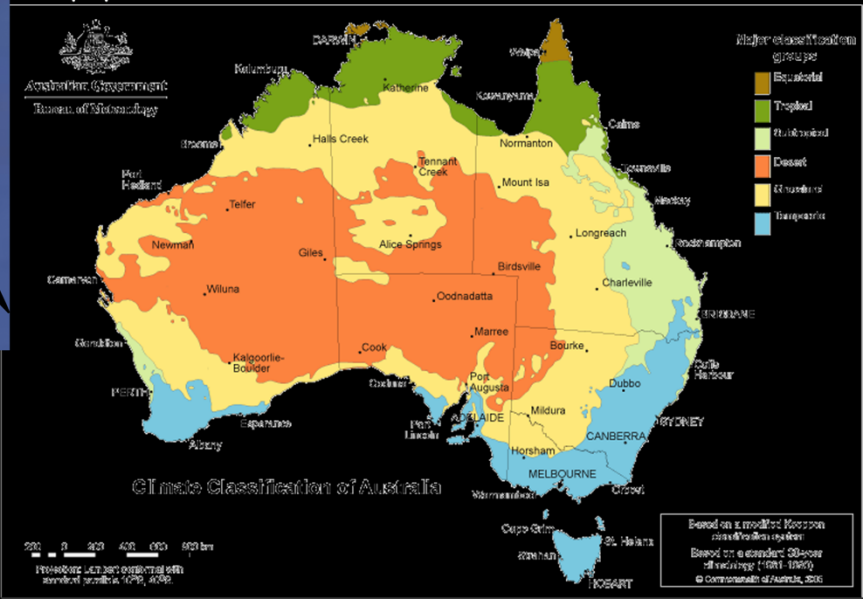
NATIONAL PARK

CRESCENT HEAD OCEAN

SOUTH KEMPSEY APPROX 22 KM NORTH WEST.

A1 - PACIFIC HIGHWAY

CLIMATE ZONE: As per the graph below Kempsey and Crescent head fall into a Sub-Tropical climate zone. Located approx a 6 hour north drive from Coffs Harbour, the Marie Short house lies. Due to the geographical location, the property experiences extreme rainfall during the wet seasons, hence Murcutt has designed for this, by building the house on stumps/posts.



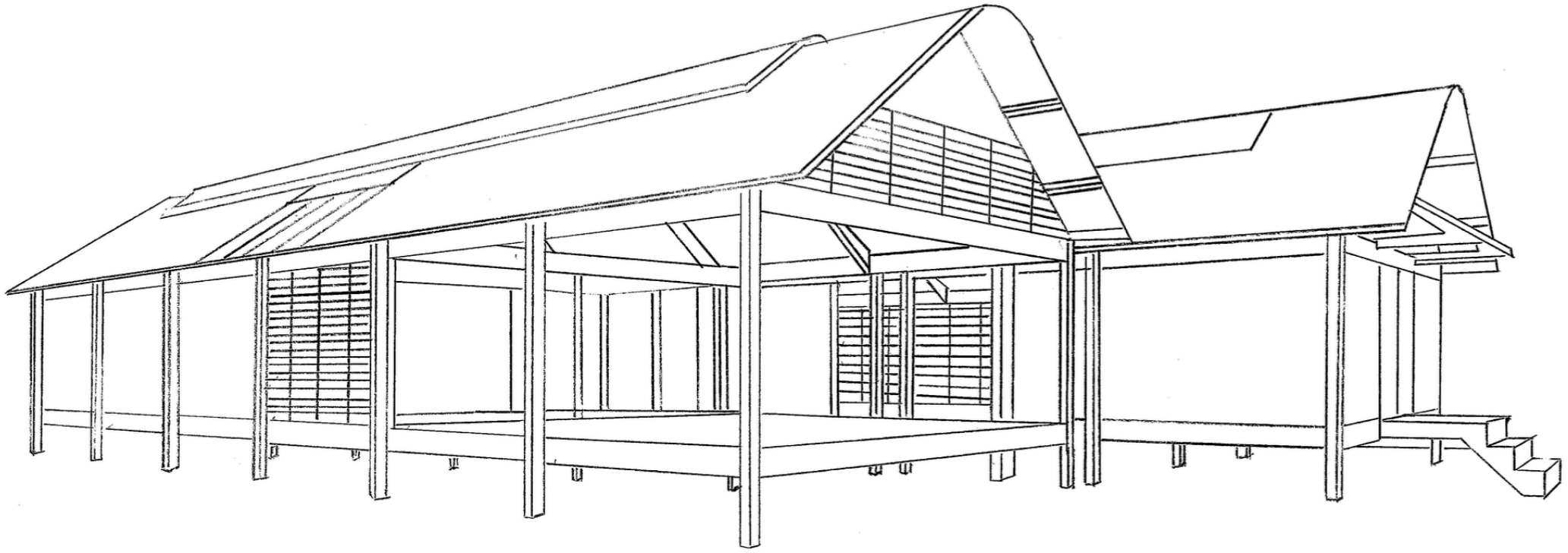
TOPOGRAPHY: Located 22km to the north the town Kempsey consists of the Maclaey River running through the centre of it out into the ocean. This river also runs through a section of the western side of the Marie Short Property. The Maria National Park is located to the north of the MS house and provides shelter to a variety of flora and fauna.

The property sits on pretty flat ground above a marshy sub-tropic flood plain, and is surrounded by an abundance of large trees and grass lands.

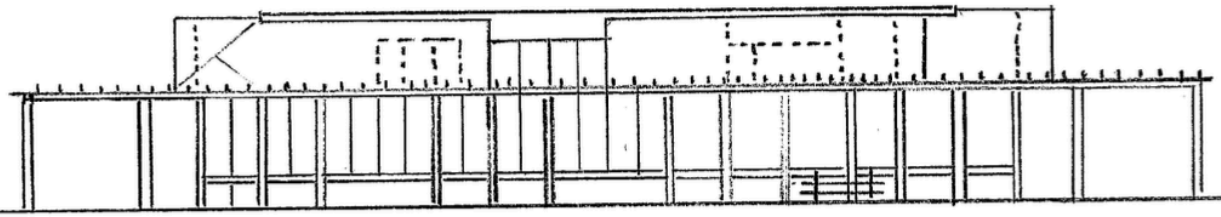
Perspective Drawings

PERSPECTIVE 1 : Marie Short House

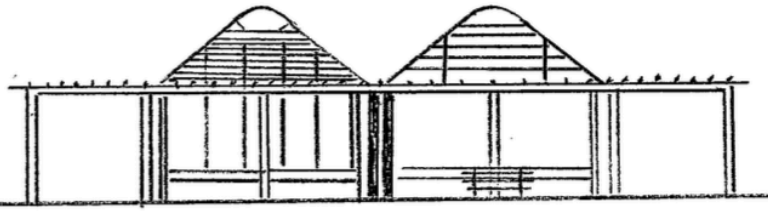
by Glenn Murcutt



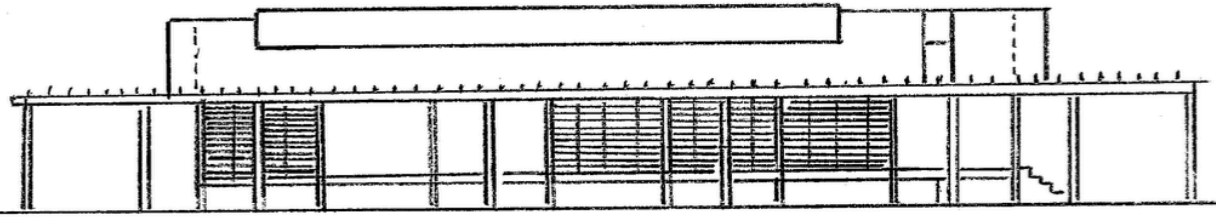
ELEVATION DRAWINGS:



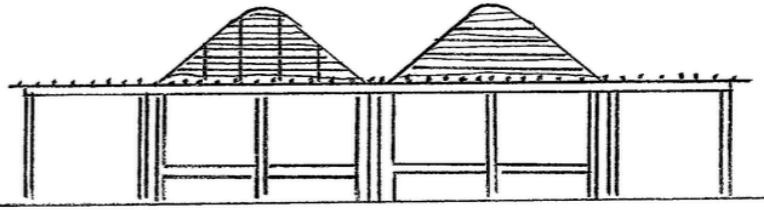
North



East

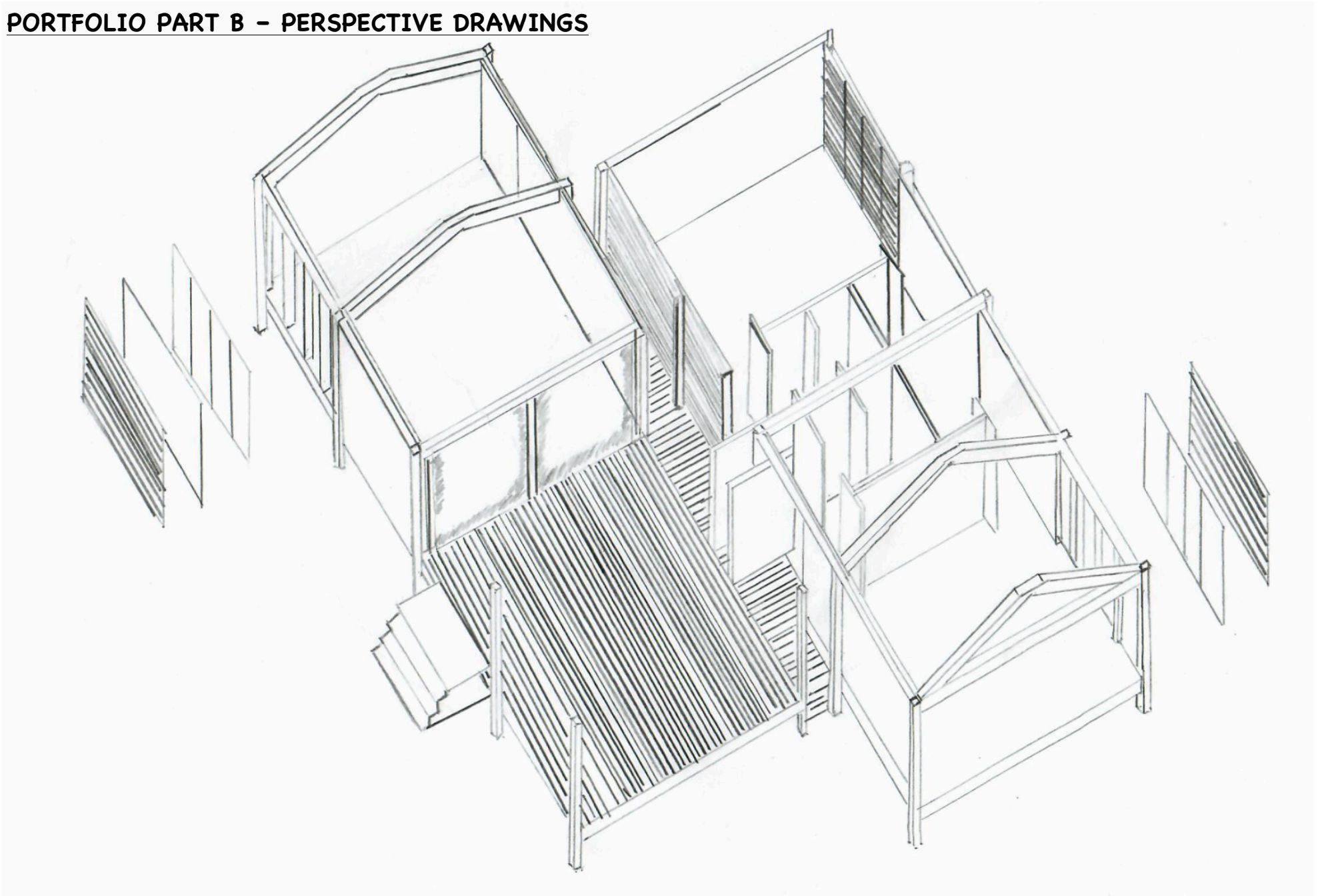


South



West

PORTFOLIO PART B – PERSPECTIVE DRAWINGS



THEME

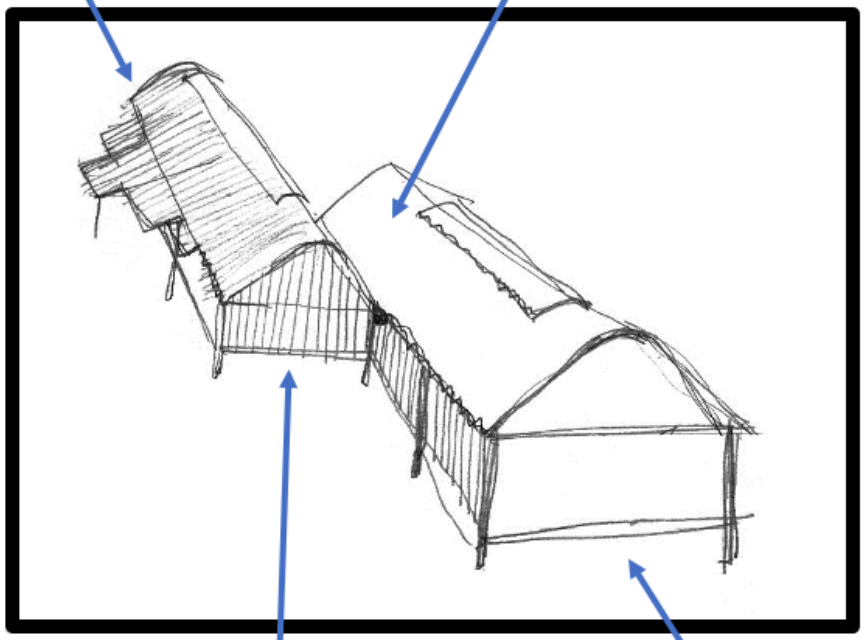
Our group has decided to base our case studies of the famous Australian architect Glenn Murcutt. Majority of his designs/houses are situated in rural farmland, forest like areas, so we have decided to choose from these houses. Glenn Murcutt motto was to **"touch the earth lightly"**, so wherever he designed a house it would be around the natural environment, rather than destroying it to make room. This motto/theme is evident in all of our case studies. In particular, with the Marie Short House, there is large bush/tree situated all around the house. But none of this vegetation was needed to move for construction of the house.

The photo shop theme we have agreed upon is having a black background (minimalistic) with the site photograph in the middle. We will continue upcoming portfolios and presentations with a similar style i.e the black simple back ground.

Model Making Plan – Marie Short House

Plywood will be used for all external walls

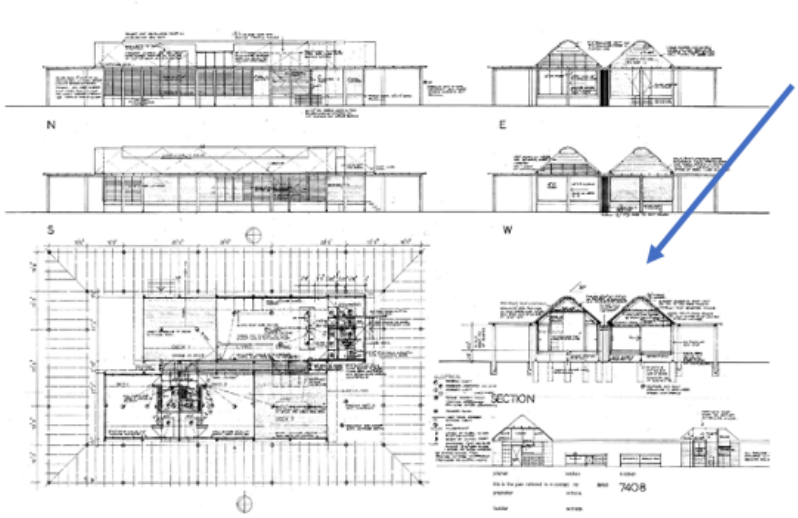
The roofing material will need to be bent in order to achieve an accurate representation. A lightweight tin or aluminum could be used to achieve this. The roofing will be separate from the walls and frame. This will allow it to be taken off/removed.



Timber louvre system – I will used left of plywood and cut them to size and stick them with super glue.

Plywood flooring to be purchased from bunnings. Approx 4mm thick.

The fireplace flu will be achieved by a steel tube from Bunnings.



The plan set the model will be based on. Can be found here: https://www.atlasofplaces.com/architecture/marie-short-house/

Budget for this model – should be around $50. I already own the power tools. So all I need is the materials from Bunnings. Materials include:

- Plywood
- Glue/super glue
- Steel tube
- Tin / Aluminum sheeting
- Level

MODEL MAKING PROCESS:

Marie Short House - Process Model (Fergus Shaw)

Steps Involved:

1. Balsa Wood, Stanley knife, hot glue gun, and timber vinyl was purchased from Bunnings
2. The flooring was cut to size and joined together with hot glue
3. The frame was erected and cut to size. The wall and lourves where than created using the balsa wood.
4. The roofing was than created. I started off doing a pitched roof. Then I soaked a thin piece of balsa wood in water and manipulated it to curve. After leaving it to rest I screwed it on with my Mikita impact driver. The Vinyl was than placed over the top to hide the screws.
5. The decking steps and internal walls where completed.

Model Making – Final Marie Short House Model

Marie Short House - Final Model (Fergus Shaw)

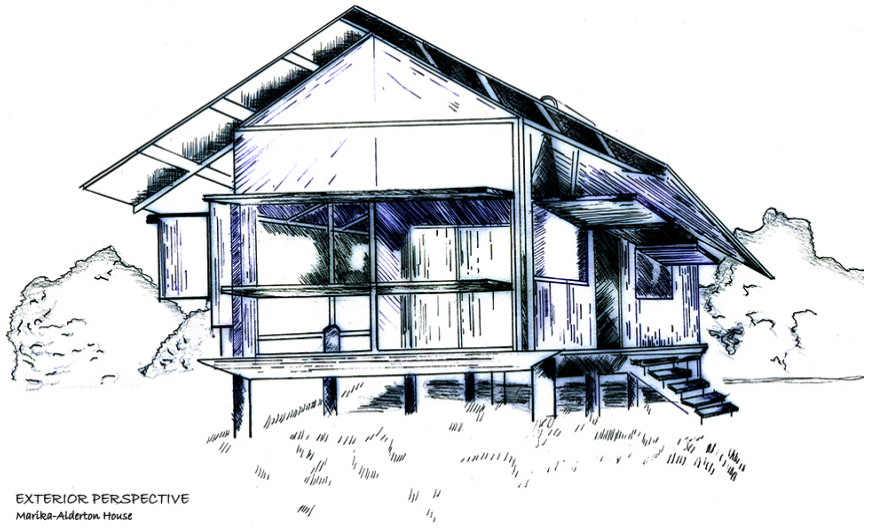
NBD1100 – Built Environment Communication & Skills

Assessment#02 – Individual Portfolio Stage 1

Palwasha Saqib

s8077065

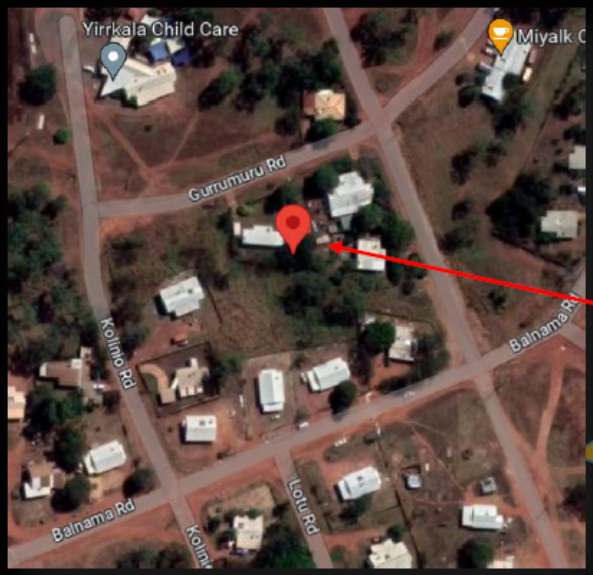
Glenn Murcutt: Marika-Alderton House



Analysis Diagram:

MARIKA-ALDERTON HOUSE

Yirrkala Community, Eastern Arnhem Land, Northern Territory, Australia



Access from Gurumuru Road & Balnama Road, with Kolino Road at the back of the property

Sea breezes from Arafura Sea

Sun path – East to West

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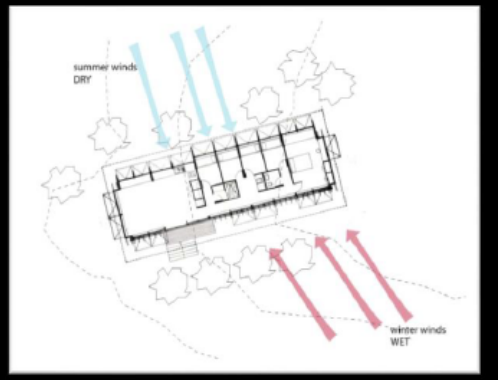
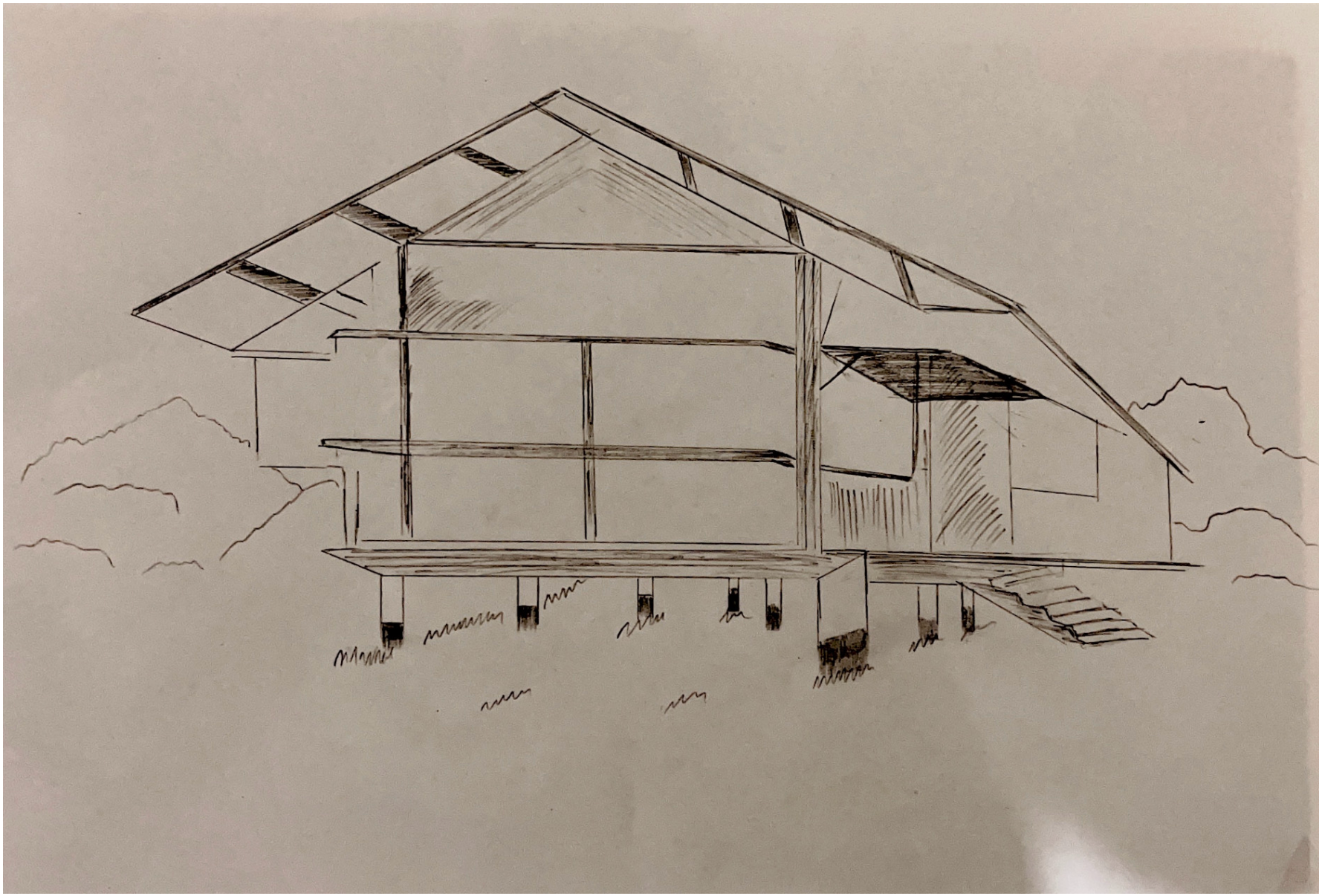
Climate Zone 1:

High humid summer and warm winter



Perspective Drawing



Model Making Plan

For physical model, I will be using recyclable moving boxes (cardboard material) and paint it red to match the exterior of the actual house. Aluminium foil shall be used to make air vents at the top and all other steel/aluminum fittings. Supporting material will be Scissors, box cutters, measuring tape, glue, paints.
All the material can be acquired from Bunnings store, wheras the box/cardboard will be re-used from existing material from house.
The maximum budget allocated for the project is $100.
Working hours allocation is 9 hours total.

Group Theme

The group is working on famous architect Glenn Murcutt's projects. His main style of architecture was making environmentally sustainable modernist houses, that responds to climate and surrounding ecology. Hence, the main theme of our group will be working on rural farmland houses, keeping the background of analysis diagrams black and the overall drawings as black and white.

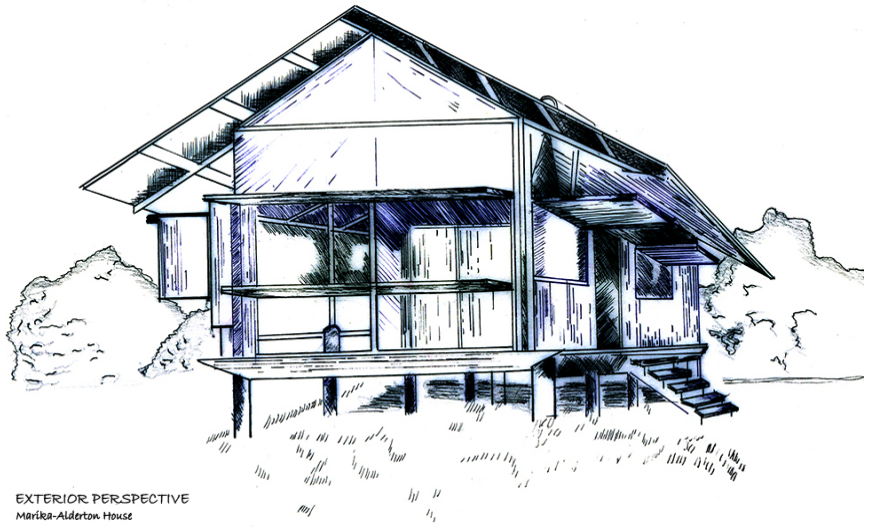
NBD1100 – Built Environment Communication & Skills

Assessment#02 – Individual Portfolio Stage 2

Palwasha Saqib

s8077065

Glenn Murcutt: Marika-Alderton House

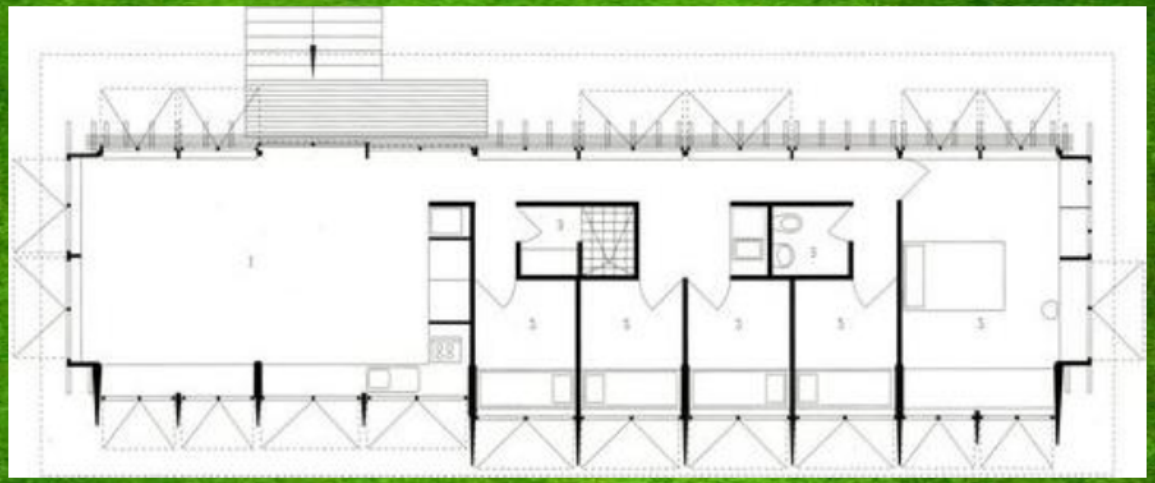


Analysis Diagram/Floor Plan:

MARIKA-ALDERTON HOUSE – FLOOR PLAN

The house is located in Australia in the Yirrkala Community, Eastern Arnhem Land, Northern Territory (Yirrkala Community, Eastern Arnheim Land, Northern Territory). To the south and west of the house is the Mangrove forest and northward the Arafura Sea, location causing winds and strong tides, with minimum temperatures of 25 and maximum of 35 degrees and high humidity.

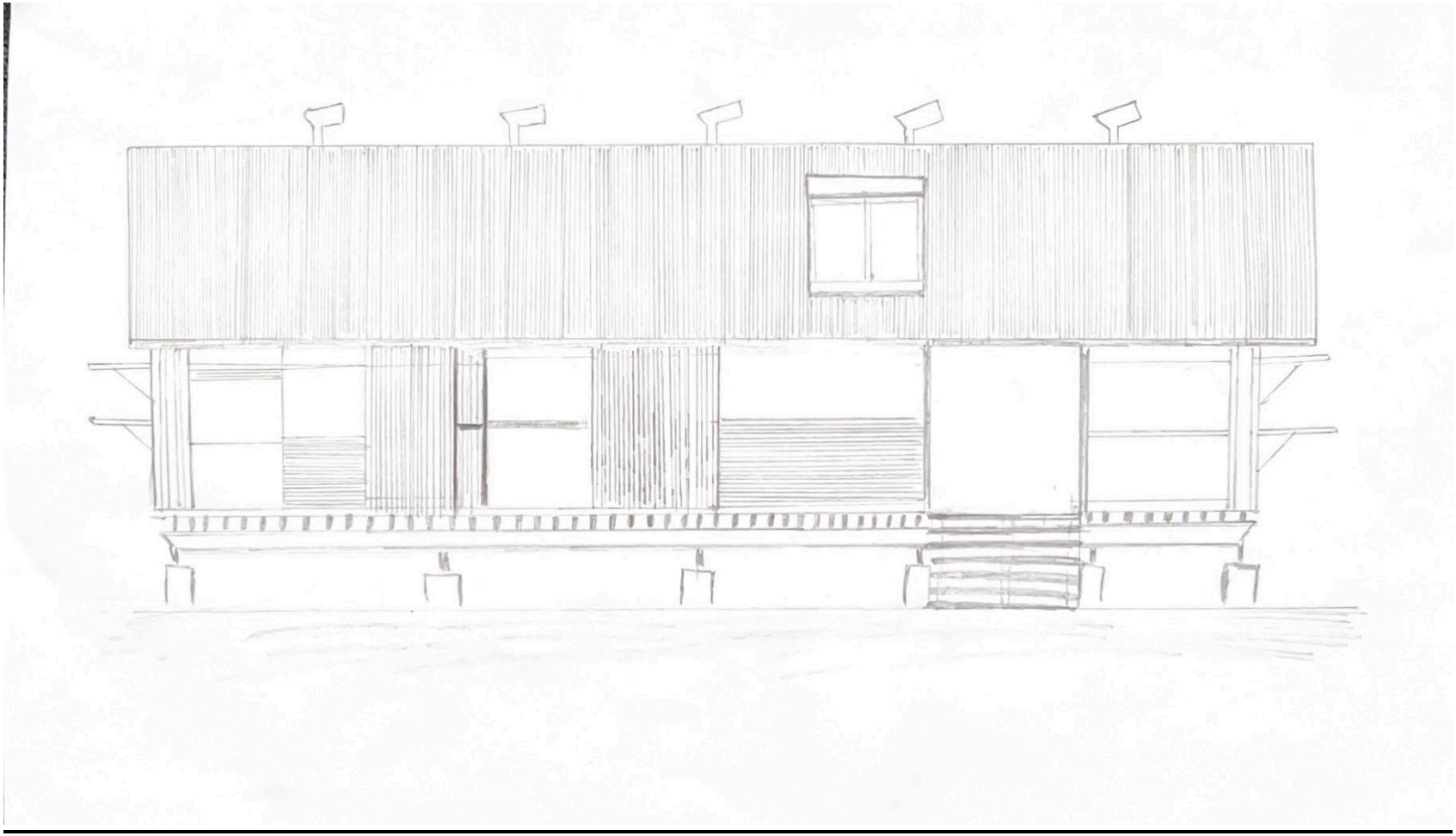
Being built in a rural Aboriginal, access to the house is by dirt roads or sea.



Access to the house is by the north, using some steps that save up on stilts.

The one-story house, elevated on stilts open in all directions and that is not a closed volume, but a shaded platform, with all kinds of racks, spoilers and brise soleil, without glass windows. Instead, the house has shutters that slope down to provide additional shade when necessary, also let in light and air flow when open. All rooms except the bathrooms enjoy all the headroom available.

Perspective Drawing



Model Making Plan – 60% completion

By the end of this assessment, 60% of the physical model has been created. Below are a few photos of work in progress:

Marina-Alderton House - Process Model (Palwasha Saqib)

Steps Involved:

1. Cardboard box, Stanley knife, heavy duty glue, aluminum foil, tooth picks
2. All the windows and sections are cut separately and joined
3. The truss is made by separating layers of cardboard.
4. Chimneys and solar panels made from aluminum foil
5. The façade is spray painted to give actual color of the house
6. Studs made from cardboard as well

Marina – Alderton House - Final Model (Palwasha Saqib)

Group Theme

The group is working on famous architect Glenn Murcutt's projects. His main style of architecture was making environmentally sustainable modernist houses, that responds to climate and surrounding ecology. Hence, the main theme of our group will be working on rural farmland houses, keeping the background of analysis diagrams black and the overall drawings as black and white.

Built Environment Communication and Skills

Assessment 4 By David Larsson s8077720

Case study: Walsh house

Walsh House

1153c Wattamolla rd Kangaroo valley

Designed by: Glenn Murcutt

Regional Analysis:

Walsh house is located in Kangaroo valley NSW.

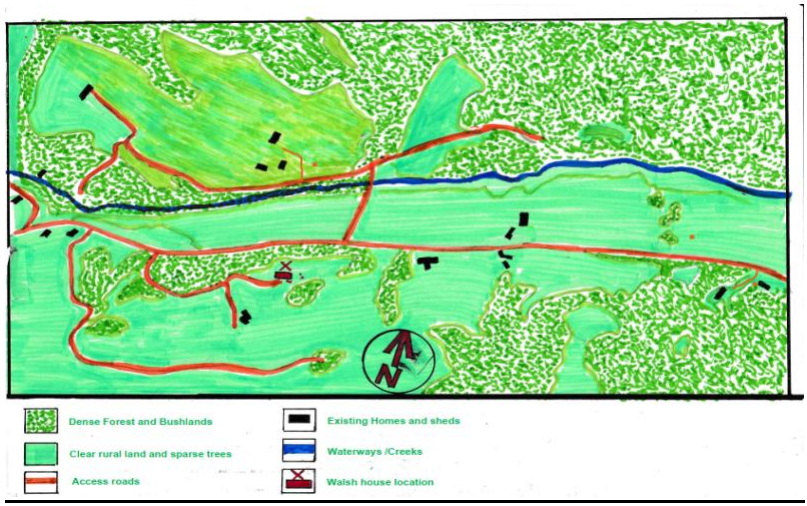
The valley is surrounded on all sides by large mountains

6kms to the west is the Kangaroo valley township village.

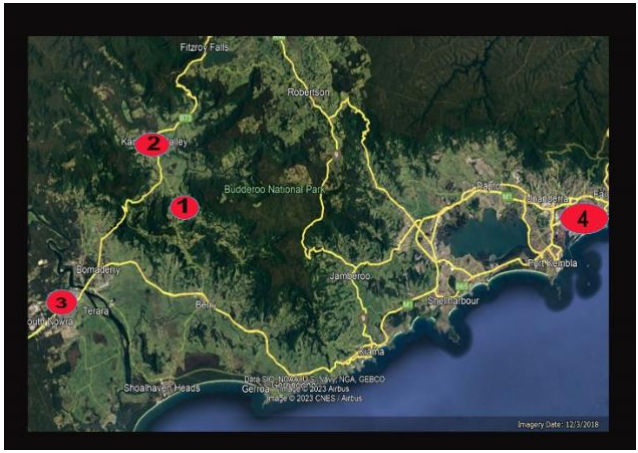
82kms to the northeast is the industrial centre of Wollongong.

23kms to the south southeast is the township of Nowra.

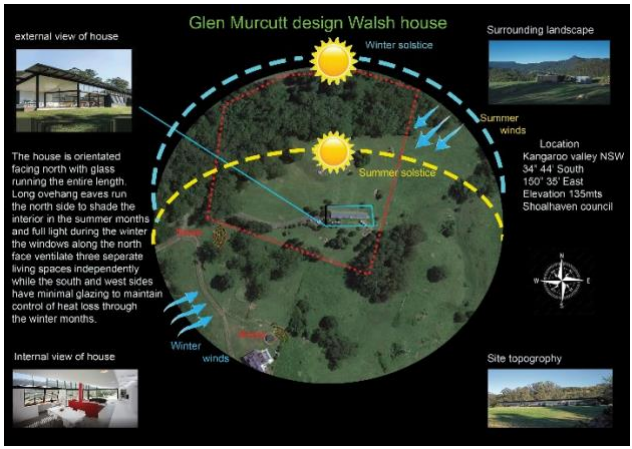
Climate zone 6



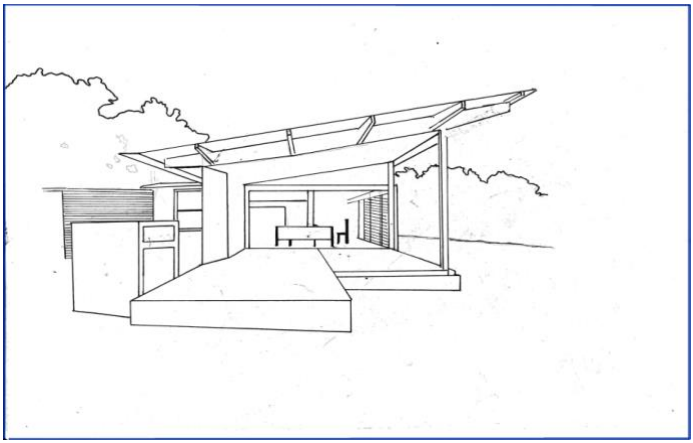
1 Walsh house 2 Kangaroo valley village 3 Nowra 4 Wollongong

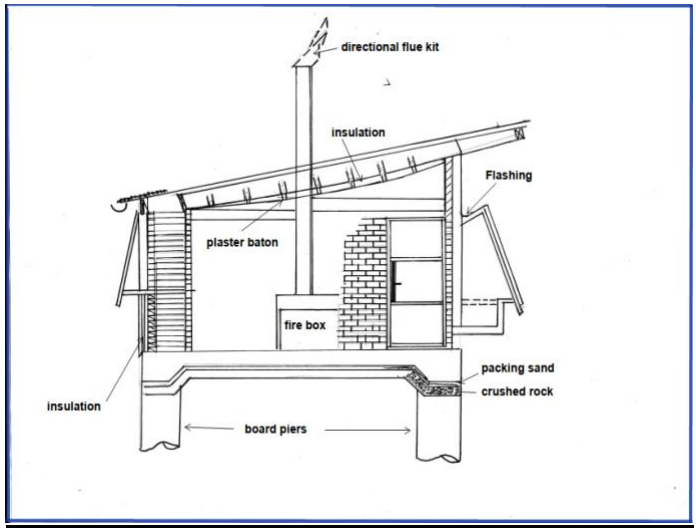


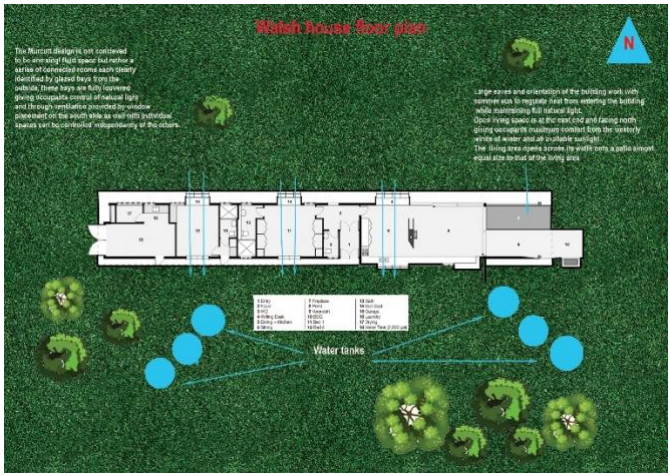
Site Analysis:



Perspective Sketches:




directional flue kit
insulation
Flashing
plaster baton
fire box
packing sand
crushed rock
insulation
board piers


Walsh house floor plan
The Nurcott design is not conceived to be one single flush space but rather a series of connected rooms each clearly identified by glazed bays from the outside. These bays are fully louvered giving occupants control of natural light and through ventilation provided by window placement on the south side as well with individual spaces can be controlled independently of the others.
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Large eaves and orientation of the building work with summer sun to regulate heat from entering the building while maximising full natural light.
Open living space is at the east end and facing north giving occupants maximum comfort from the westerly winds of winter and all available sunlight.
The living area opens across the width onto a patio almost equal size to that of the living area
Water tanks

Model Process

Finished Model